GA PSV EXAMPLE (X-BOW-LIKE)

92 m diesel-electric platform supply vessel with an inverted, X-Bow-like bow — educational example

Overview

A large platform supply vessel for offshore logistics, with forward accommodation and wheelhouse, a long open cargo deck aft, and dry bulk, liquid mud, brine, fuel, water and special liquid tanks below deck. The arrangement is simplified from a public PSV data sheet; for this example the bow has been redrawn as an inverted, X-Bow-like bow, with its most forward point low down and the stem leaning back towards the accommodation. Diesel-electric machinery drives two azimuth propellers aft; bow thrusters and DP2 positioning give station keeping.

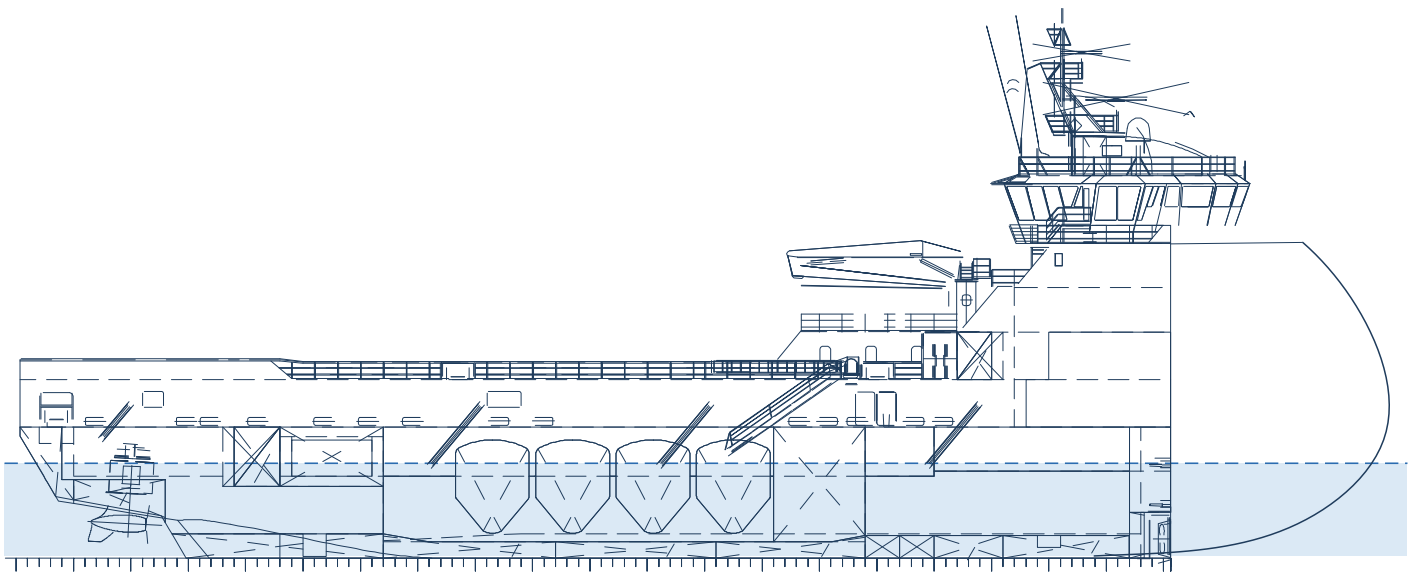
Principal particulars	
Length overall	92 m
Length between perp.	85 m
Breadth, moulded	20 m
Depth to main deck	8.2 m
Design draught	6.2 m
Deadweight	4,900 t
Clear deck	58 m x 17 m
Service speed	14.5 kn
Complement	28 persons

Tank capacities (rounded)	
Fuel oil	1,450 m^3
Fresh water	1,000 m^3
Ballast / drill water	1,700 m^3
Liquid mud	1,300 m^3
Brine	1,300 m^3
Dry bulk (4 tanks)	250 m^3

Machinery and propulsion	
Power plant	Diesel-electric, 690 V / 60 Hz, four generator sets
Propulsion	2 x azimuth pulling propellers, about 2,200 kW each
Bow	Tunnel thruster(s) + retractable azimuth
Positioning	DP2, two DGPS + optical reference

Cargo and deck	
Dry bulk	4 cement / barite tanks, pneumatic
Liquid cargo	10 round mud / brine tanks; LFL and base oil
Deck crane	Knuckle boom, 3 t at 18 m
Winches	Anchor / mooring winch, 2 tuggers, 2 capstans
Lifesaving	MOB boat, 2 x 25-person rafts

PROFILE



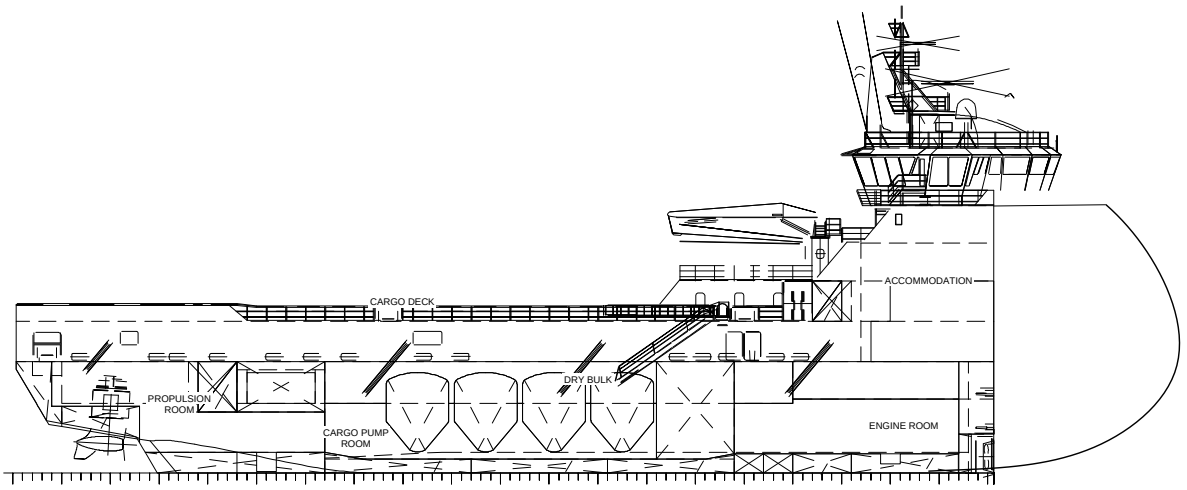
Traced profile with the X-Bow-like bow (water below the design waterline in blue). Full arrangement on page 2.

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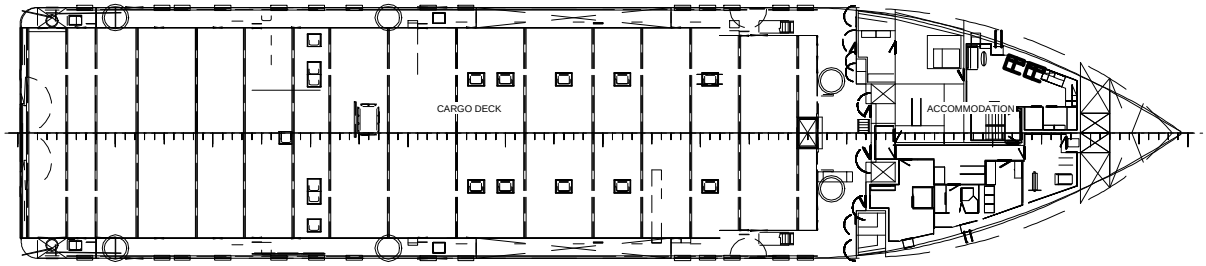
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GENERAL ARRANGEMENT

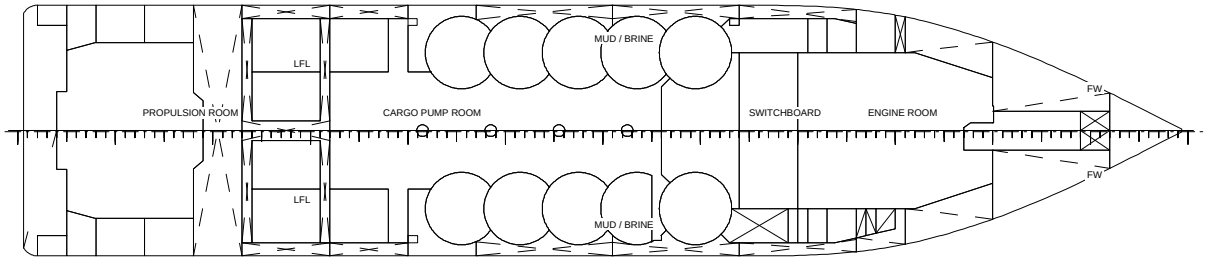
PSV EXAMPLE (X-BOW-LIKE)
Simplified educational drawing — not to scale for use



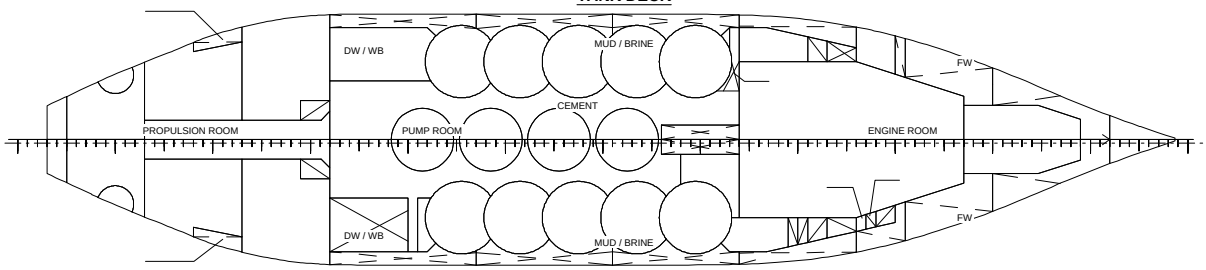
PROFILE



MAIN DECK



TANK DECK



TANK TOP



MAIN DIMENSIONS	
Length o.a.	92 m
Length b.p.	85 m
Breadth moulded	20 m
Depth to main deck	8.2 m
Design draught	6.2 m
Deadweight	4,900 t
Clear deck	58 x 17 m

CAPACITIES (rounded)	
Fuel oil	1,450 m3
Fresh water	1,000 m3
Ballast / drill water	1,700 m3
Liquid mud	1,300 m3
Brine	1,300 m3
Dry bulk	250 m3

MACHINERY	
Plant	diesel-electric
Propulsion	2 x azimuth
Bow	tunnel + retractable
Positioning	DP2
Speed	14.5 kn

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